

Therefore, for t sufficiently large,

$$(4.12) \quad \begin{aligned} & \exp(-tP)(1 - \Pi_N)u(z) \\ &= \sum_{\alpha \in S_{N+1}} \frac{C_\alpha}{((N+1)(R(M)-1))!} A(t)^{N+1} e^{-it\Im\lambda_\alpha} \frac{\partial^\alpha u(0)}{\alpha!} z^{\tilde{\alpha}} \\ & \quad + \mathcal{O}(t^{-1}A(t)^{N+1}\|u\|) \end{aligned}$$

Then (4.9) follows from (4.11) and the triangle inequality.

The claim about weak convergence comes from the observation that, if $\exp(-t\tilde{P})$ converges weakly, then

$$\#\{\Im\lambda_\alpha : \alpha \in S_{N+1}\} = 1;$$

otherwise, $e^{iat}A(t)^{-N-1}$ contains oscillating factors. Since this set can be expressed as the collection of sums (allowing repetition) of $N+1$ imaginary parts $\Im\lambda_j$ where $\Re\lambda_j = \rho(M)$ and $\tilde{r}_j = R(M)-1$, this collection consists of one value if and only if

$$\#\{\Im\lambda_j : \Re\lambda_j = \rho(M) \text{ \& } \tilde{r}_j = R(M)-1\} = 1.$$

Note that, even if this is true, the eigenvalue $\rho(M) + i\Im\lambda_j$ could correspond to many Jordan blocks of the same size.

In the case that there is only one such $a = \Im\lambda_\alpha$, it is clear from (4.12) that

$$\begin{aligned} & e^{iat}A(t)^{-N-1} \exp(-t\tilde{P})(1 - \Pi_N)u(z) \\ &= \sum_{\alpha \in S_{N+1}} \frac{C_\alpha}{((N+1)(R(M)-1))!} \frac{\partial^\alpha u(0)}{\alpha!} z^{\tilde{\alpha}} + \mathcal{O}(t^{-1}\|u\|), \end{aligned}$$

proving convergence in operator norm. Again, these statements for $\tilde{P} = \mathcal{V}_{\tilde{G}}^* P \mathcal{V}_{\tilde{G}}$ lead immediately to the corresponding statements for P , and so the proposition is proven. $\square$

Remark 4.5. The projections

$$u(z) \mapsto \frac{\partial^\alpha u(0)}{\alpha!} z^\alpha$$

can be seen to be bounded on H_Φ for the same reasons that each Π_N from (4.1) is a bounded projection. A more detailed analysis is carried out in [32]; while we recall that the norms of these rank-one projections must be bounded by $Ce^{C|\alpha|}$ for some $C > 0$ depending on Φ , we do not pursue this question here.

Remark 4.6. When $\text{Spec } M \subset \{\Re\lambda > 0\}$, we have from the case $N = 0$ of Proposition 4.4 that, for t sufficiently large,

$$\exp(-tP)u(z) = u(0) + \mathcal{O}(t^{R(M)-1}e^{-t\rho(M)}\|u\|),$$

with error a function in H_Φ ; furthermore, the error estimate is sharp. There is therefore a large gap between (2.34), which has exponential growth as an upper bound as $\tau = -t \rightarrow -\infty$, and the true behavior which is bounded with an exponentially small error.

This gap is explained under the hypotheses of Section 4.3 where the value of the norm of $\|\exp(-tP)\|$ is known exactly, but the question remains open in the

general case. Because $u(z) \mapsto u(0)$ is an orthogonal projection only when the harmonic part $\Re h(z)$ of Φ from (2.9) vanishes (see Proposition 4.9), we remark that if $\text{Spec } M \subset \{\Re \lambda > 0\}$, then

$$\lim_{t \rightarrow \infty} \|\exp(-tP)\| = 1 \iff h = 0.$$

4.2. Weak ellipticity and small-time regularization. It is already apparent from Theorem 2.9 that the rate of change of $\Phi(e^{tM}z) - \Phi(z)$ as a function of t plays an important role in behavior of the solution operator for small times. We begin by identifying that rate under the (non-strict) ellipticity hypothesis (2.38).

Theorem 4.7. *Let Φ satisfy (2.1). Assume that the non-strict ellipticity condition (2.38) holds for a matrix M and fix $z \in \mathbb{C}^n$. Using the notation (2.39), let*

$$(4.13) \quad I = I(z) = \min \{k \geq 0 : \Theta(M^k z) \neq 0\}.$$

Then either $I \leq 2n - 2$ and, as $t \rightarrow 0$,

$$(4.14) \quad \Phi(e^{tM}z) - \Phi(z) = \frac{1}{(2I+1)!} \binom{2I}{I} \Theta(M^I z) t^{2I+1} + \mathcal{O}(t^{2I+2})|z|^2,$$

or $I = \infty$ and

$$(4.15) \quad \Phi(e^{tM}z) = \Phi(z), \quad \forall t \in \mathbb{R}.$$

Remark. When $I(z) = \infty$ for some $z \in \mathbb{C}^n \setminus \{0\}$, we conclude that $\exp(-tP)$ is never compact for any $t \in \mathbb{R}$ by Theorem 2.9.

Proof. Regarding Θ as a quadratic form in $2n$ real variables, we have that Θ is positive semidefinite by our assumption (2.38) and therefore its zero set coincides with the kernel of its Hessian matrix. This is a linear condition, so by the Cayley-Hamilton theorem we have that if $I \geq 2n - 1$ then $I = \infty$ for I in (4.13).

To analyze derivatives of $\Phi(e^{tM}z)$, particularly of higher order, it is convenient to associate $\Phi : \mathbb{C}^n \rightarrow \mathbb{R}$ with a natural real-valued real-bilinear form acting on $\mathbb{R}^{2n}$. That is, let

$$(4.16) \quad \Phi(z, \zeta) = \Re(z \cdot \Phi'_z(\zeta))$$

denote the unique symmetric real-bilinear form on $\mathbb{C}^{2n}$ such that $\Phi(z, z) = \Phi(z)$. Then we compute that

$$(4.17) \quad \frac{d^k}{dt^k} \Phi(e^{tM}z) = \sum_{j=0}^k \binom{k}{j} \Phi(M^j e^{tM}z, M^{k-j} e^{tM}z).$$

It is also useful to similarly extend Θ , which is here a positive semi-definite real-valued real-quadratic form thanks to (2.38), to a real-valued real-bilinear form. We note that $\Theta(z) = 2\Phi(Mz, z)$ and therefore we may express the extension of Θ in terms of that of Φ :

$$(4.18) \quad \Theta(z, \zeta) = \Phi(Mz, \zeta) + \Phi(z, M\zeta).$$

To establish the theorem, we show that, for $k \in \mathbb{N}$,

$$(4.19) \quad \left. \frac{d^j}{dt^j} \Phi(e^{tM}z) \right|_{t=0} = 0, \quad \forall j = 1, \dots, 2k+1$$